# Introduction to Economics

Class 7 — Elasticity

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Universitat de Barcelona · Group A6

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## Slide 1 Title

Class 6 settled which way a price moves when a curve shifts; Class 7 asks how much. One number does the work: elasticity, the percentage change in quantity over the percentage change in price. The first hour defines it, computes it between two points and along a line, and uses it to say what a price change does to a seller's money. The second hour pays off the opening puzzle, adds income and cross-price elasticities, and gives sellers an elasticity of their own. The thread is a real pair of numbers: Spain sent a third less olive oil abroad in 2023 and got the same money back.

# Quiz 4

10 minutes · pens down at 10:10

Circle the letter of your answer in the block at the top.

Changed your mind? Cross out that circle and circle the new letter.

Write your name on the block, tear the strip off along the dashed line, hand in *only the strip*.



## Quiz Start of class, on paper

A short quiz on paper opened the class: circle the letter of your answer in the block at the top of the sheet, write your name on the block, tear the strip off along the dashed line and hand in only the strip.

## Plan for today

| | |
|---|---|
| 10:00 | Quiz 4 |
| 10:10 | Warm-up; Class 6, one more look |
| 10:18 | Olive oil abroad: a third less, the same money |
| 10:21 | Elasticity: the number, the midpoint |
| 10:28 | One line, many elasticities; in pairs |
| 10:38 | What makes demand elastic |
| 10:43 | Price × quantity: the box |
| 10:50 | Break |
| 11:00 | Topic fixed; Problem set 1 out |
| 11:02 | The olive oil answered; raise or cut? |
| 11:14 | Income and cross-price elasticities |
| 11:27 | Supply: when sellers can respond |
| 11:38 | In one line; before Monday 19 October |



**Slide 2** Plan for today

The class at a glance. A short warm-up on two moves from Classes 5 and 6, then a last look at Class 6's teaser, then a real pair of numbers that only today's idea explains. The definition of elasticity, the method for computing it between two points, and a line that is elastic at one end and inelastic at the other; then what a price change does to the money in a market. After the break the puzzle gets its answer, two more elasticities whose sign is the point, and the sellers' side, where time is the key.

Two halves of fifty minutes with a ten-minute break between them.

## Warm-up: which curve moves, and which way?

### Coach tickets, Barcelona–Valencia: for students, an inferior good

1. Students' incomes fall
   **demand shifts right: an inferior good**
2. Car-sharing rides get dearer
   **demand shifts right: a substitute got dearer**
3. The coach fare falls
   **no shift: more tickets, along the demand curve**
4. Diesel gets dearer
   **supply shifts left; the fare rises along demand**



**Slide 3** Warm-up: which curve moves, and which way?

A warm-up on the move that is easiest to get wrong: telling a shift of a curve from a move along it. A change in the good's own price moves buyers along its demand curve; anything else that changes what buyers want at every price shifts demand; a change in costs shifts supply. For students the coach is an inferior good, so falling incomes shift its demand right, and dearer car-sharing, a substitute, does the same.

A cheaper coach fare shifts nothing: more tickets are bought along the same curve. Dearer diesel is a cost, so supply shifts left; the fare rises and buyers move along their demand curve to fewer tickets, with no second shift of demand.

## Warm-up: what does the whole market buy?

**Padel-court hours a month**

| price | Ana | Biel | the market |
|---|---|---|---|
| 10 euros | 5 | 3 | $5 + 3 = 8$ |
| 20 euros | 3 | 1 | $3 + 1 = 4$ |

Carla joins: 2 hours at 10 euros, 1 at 20. The market curve?

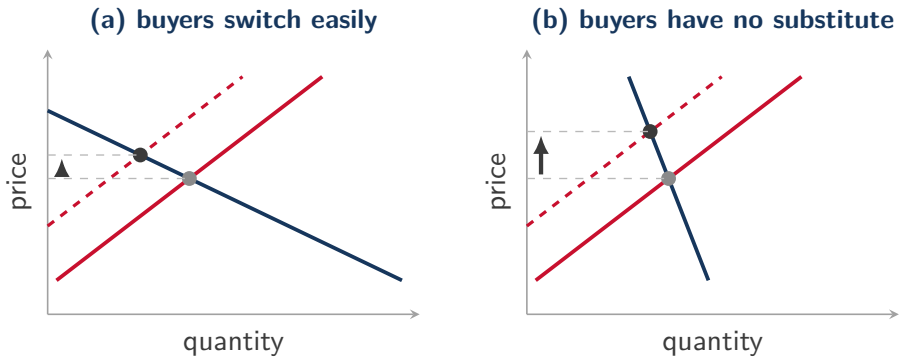
**shifts right: 10 hours at 10 euros, 5 at 20**



**Slide 4** Warm-up: what does the whole market buy?

The market's demand adds up what every buyer wants at the same price: at 10 euros Ana's 5 hours and Biel's 3 make 8, at 20 euros 3 and 1 make 4. The curves are added sideways, quantities at one price, never prices at one quantity, and the market counts every buyer, not the average one. A new buyer adds hours at every price, so the market curve shifts right: the number of buyers is one of the things that shift demand.

## Class 6 again: the same shift — where does the price jump more?




**(b).** Same shift; the price jumps more where buyers cannot walk away.



**Slide 5** Class 6 again: the same shift, where does the price jump more?

Class 6's last picture, as a question. The same supply shift in two markets: in (a) buyers switch easily and the demand line is flat; in (b) they have no substitute and it is steep. The answer is (b), and the reason is that the shift is identical, so the whole difference sits on the buyers' side. Where buyers cannot walk away, the price has to do the adjusting; where they can, the quantity does.

Class 6 said steep and flat. Today those words become a number, so that two markets can be compared without a drawing.

## Olive oil abroad in 2023: a third less oil, the same money

1.07 million tonnes → **0.68 million tonnes**

shipped abroad, 2022 → 2023

4.10 bn € → **4.15 bn €**

what came back, 2022 → 2023

**What does that say about the buyers abroad?**

Spain's olive-oil exports, customs data (AEAT), in MAPA's 2023 product report; 2023 provisional.



**Slide 6** Olive oil abroad in 2023: a third less oil, the same money

The hook, from the customs office. Spain shipped 1.07 million tonnes of olive oil abroad in 2022 and 0.68 million in 2023, after the dry year of Class 6: about a third less. The money that came back was 4.10 billion euros, then 4.15 billion: the same. A third less of something and the same money is odd only until you ask how the buyers abroad reacted to the price. The answer waits for the second half; the slide is here so that the question is in your head while the tool is built.

## Elasticity: how much the quantity moves, in percentages

$$\text{price elasticity of demand} = \left| \frac{\%\text{ change in quantity demanded}}{\%\text{ change in price}} \right|$$

$> 1$ **elastic**: the quantity moves more than the price

$< 1$ **inelastic**: the quantity moves less than the price

$= 1$ **unit elastic**: the same

Percentages, so cups and tonnes compare. The absolute value drops the minus sign.



**Slide 7** Elasticity: how much the quantity moves, in percentages

The definition in words: the price elasticity of demand is the percentage change in the quantity demanded divided by the percentage change in the price. Percentages, because forty fewer cups and 0.4 million tonnes less cannot be compared, while two percentages can. Along a demand curve the quantity falls when the price rises, so the ratio is negative; so the definition takes its absolute value: the sign is dropped and only the size is read.

Elastic means stretchy: a small pull on the price and the quantity moves a lot, more than one per cent for each per cent of price. Inelastic is the reverse; unit elastic is the border, where the two percentages are equal.

## Two points, one number: percentages of the midpoint

The neighbourhood's cups: 2.00 → 2.50, 160 → 120 a day

| computed from | price | cups | elasticity |
|---|---|---|---|
| the start, 2.00 | +25 % | −25 % | 1.0 |
| the start, 2.50 | −20 % | +33 % | 1.67 |
| **the midpoint** | **+22 %** | **−29 %** | **1.29** |

$$\left| \frac{(Q_2 - Q_1) \,/\, \tfrac{1}{2}(Q_1 + Q_2)}{(P_2 - P_1) \,/\, \tfrac{1}{2}(P_1 + P_2)} \right|$$

Divide each change by the average of the two values, not by the start.

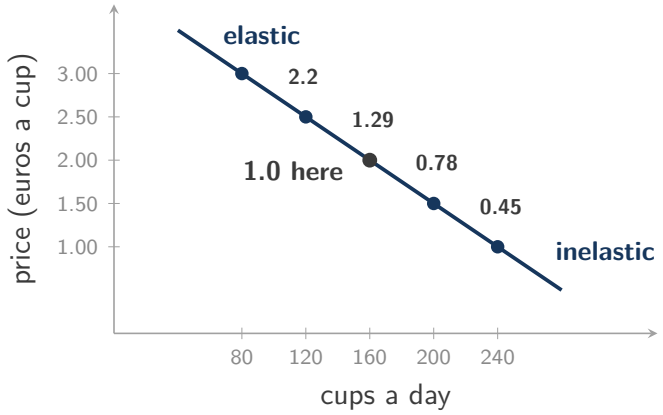


**Slide 8** Two points, one number: percentages of the midpoint

The same two points of the coffee table, 2.00 with 160 cups and 2.50 with 120, give two different elasticities when the percentages are taken from the start: 1.0 going up, 1.67 going down, because a percentage depends on its base. The midpoint method removes the ambiguity: divide each change by the average of the two values. Here 0.50 over 2.25 and 40 over 140, so 22 and 29 per cent, and the elasticity is 1.29 in either direction.

In this course an elasticity between two points always means the midpoint method; the problem sets use it.

## One straight line: elastic at the top, inelastic at the bottom




Same slope everywhere; the percentages change:
few cups and a high price at the top, many cups and a low price at the bottom.



**Slide 9** One straight line: elastic at the top, inelastic at the bottom

The whole coffee table on its line. Every step of 0.50 in price costs 40 cups, the same slope everywhere, yet the elasticity runs from 0.45 at the bottom to 2.2 at the top. What changes is the base of the percentages: low on the line, 40 cups is a small share of many cups and 0.50 a large share of a low price, so the quantity moves less than the price; high on the line it is the reverse. Exactly in the middle, 160 cups at 2.00, the elasticity is 1.

So a single line does not have an elasticity; it has every elasticity. "Steep means inelastic" is a rule for comparing two lines through one point, not for reading one line.

## In pairs, four minutes: compute and classify

### Elastic, inelastic, or unit elastic?

1. Bikinis at the bars: 5.00 → 5.50 euros, 200 → 150 a day
   **3.0, elastic**
2. A single metro fare +10 %; rush-hour journeys −2 %
   **0.2, inelastic**
3. The zoo's ticket +20 %; visitors −20 %
   **1.0, unit elastic**



## Demand is elastic when buyers can walk away

- Close substitutes nearby: Bar Nou, or the bar next door
- A luxury, not a necessity: a dinner out, not bread
- A narrow market: one bar's cup, not "coffee"
- Time to adjust: a year, not a week
- A big slice of the budget: rent, not chewing gum



## Slide 10 In pairs, four minutes: compute and classify

Exercise 1. Item 1 is Class 6's bikini table: a fall of 50 on an average of 175 is 29 per cent, a rise of 0.50 on an average of 5.25 is 9.5 per cent, so the elasticity is 3.0, elastic, as a snack with a substitute on every corner should be. Item 2 gives the percentages already, so no midpoint is needed: 2 over 10 is 0.2, inelastic; at rush hour people have to get to work.

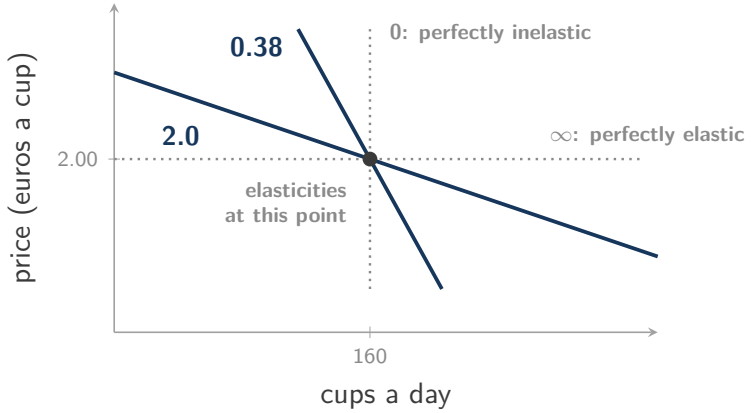
Item 3 also gives the percentages: a 20 per cent rise in the ticket price and a 20 per cent fall in visitors, so 20 over 20 is 1.0, unit elastic. The price rise and the fall in visitors exactly offset each other; slide 15 shows what that means for the zoo's takings.

## Slide 11 Demand is elastic when buyers can walk away

The reading's five determinants in one sentence: demand is elastic when buyers can walk away, and walking away is easy when there is somewhere to go. Close substitutes are somewhere to go; a necessity such as bread or the morning metro gives nowhere. The narrower the market, the more substitutes lie outside it: one bar's cup is elastic, coffee less so, hot drinks less still. Time adds options: in a week you keep your commute, in a year you can change it. A big slice of the budget makes you look around; a small one does not.

The common error is to think that expensive goods are inelastic. What matters is the alternatives and the share of the budget, not the price tag.

## Through one point: the flatter the line, the more elastic




Same axes only. A line that looks flat on a stretched axis means nothing.

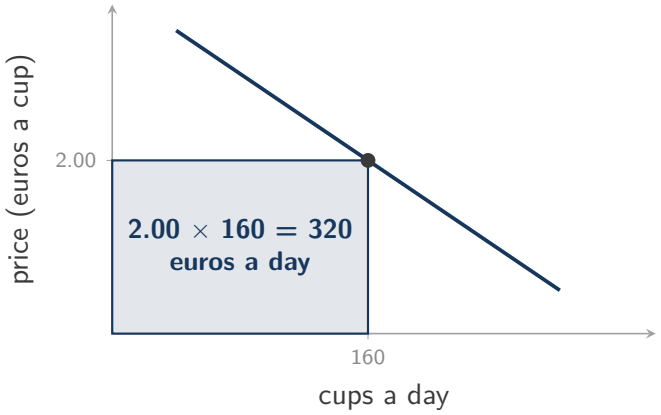


**Slide 12** Through one point: the flatter the line, the more elastic

Two demand curves through the same point on the same axes. The flat one has elasticity 2.0 there, the steep one 0.38: the same price rise costs more than five times as many cups on the flat line. The extremes have names: vertical, the quantity never moves, perfectly inelastic, elasticity zero; horizontal, perfectly elastic, buyers take any amount at one price and none above it, Class 5's price taker seen from the seller's side.

The warning is the point of the slide: the comparison works only through one point on the same axes. Stretch the price axis and any line looks flat. Read the number, not the look.

## What buyers spend is what sellers get: price × quantity, a box




**Total expenditure** of buyers = **total revenue** of sellers: one box, two names.

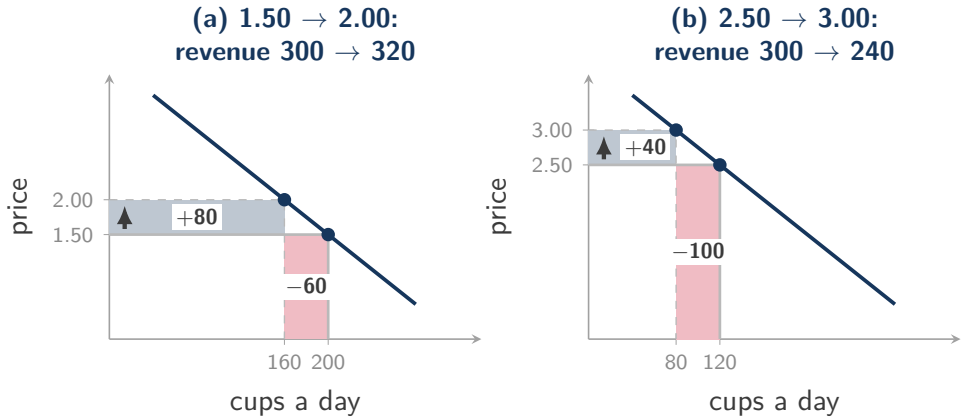


**Slide 13** What buyers spend is what sellers get: price × quantity, a box

The money in a market is the price times the quantity: 2.00 times 160 cups is 320 euros a day, the box under the point on the demand curve, as tall as the price and as wide as the quantity. Buyers call that box their spending and sellers their revenue; it is one box with two names. Revenue is not profit: the beans and the barista are paid out of it later.

The slide sets up the bar owner's question: raise the price and the box gets taller but narrower. Which wins is the next slide.

## A dearer cup: more money at the bottom of the line, less at the top




Inelastic stretch (0.78): the gained strip wins. Elastic stretch (2.2): the lost one does.



**Slide 14** A dearer cup: more money at the bottom of the line, less at the top

Two price rises of 0.50 on the same line. In (a), low on the line where the elasticity is 0.78, the bars gain 0.50 on each of the 160 cups they still sell and lose 40 cups that were worth 1.50 each: the gained strip beats the lost one and revenue rises from 300 to 320. In (b), high on the line where the elasticity is 2.2, the gain is 0.50 on only 80 cups while the 40 lost cups were worth 2.50 each: the lost strip wins and revenue falls from 300 to 240.

Same slope, same 0.50, opposite answers, because the cups that walk away were worth different amounts. The box is biggest in the middle, 320 at 2.00, exactly where the elasticity is 1.

## Raise the price when demand is inelastic; cut it when elastic

| demand | price up | price down |
|---|---|---|
| **inelastic** ($< 1$) | revenue up | revenue down |
| **unit** ($= 1$) | unchanged | unchanged |
| **elastic** ($> 1$) | revenue down | revenue up |

Inelastic: revenue follows the price. Elastic: it runs the other way.



**Slide 15** Raise the price when demand is inelastic; cut it when elastic

The last two slides as a table. When demand is inelastic, price and revenue move together: a rise brings more money, a cut less. When demand is elastic they move apart: a rise loses money, a cut brings more. At unit elasticity the box does not change.

For a seller who sets a price, a metro authority, a cinema, a bar, this is the first thing to know. For a whole market of price takers, like the olive-oil growers of Class 6, the same rule says what a smaller harvest does to the sector's money, which is the puzzle waiting for the second half.

## Which curve moves, and how many members? (optional)

### Gym memberships in a city; price elasticity of demand 1.5

1. Memberships get 10 % dearer
   **no shift: along demand, members −15 %**
2. Workout apps get dearer
   **demand shifts right: a substitute**
3. Gyms' electricity bills double
   **supply shifts left; price up, members down along demand**



**Slide 16** Which curve moves, and how many members? (optional)

An optional exercise joining today's number to Class 5's distinction. A 10 per cent rise in the membership price is the good's own price: no shift, a move along demand, and with an elasticity of 1.5 the members fall by 15 per cent (revenue falls too, since demand is elastic). Dearer workout apps are a dearer substitute, so the demand for gyms shifts right. Dearer electricity is a cost: supply shifts left, the price rises and buyers move along their demand curve; with elastic demand much of the change shows up as fewer members rather than a much higher price.

## Break

Back at 11:00



**Slide 17** Break

Ten minutes. After it: the project and Problem set 1, then the olive oil gets its number.

## Today: your topic is fixed; Problem set 1 is out

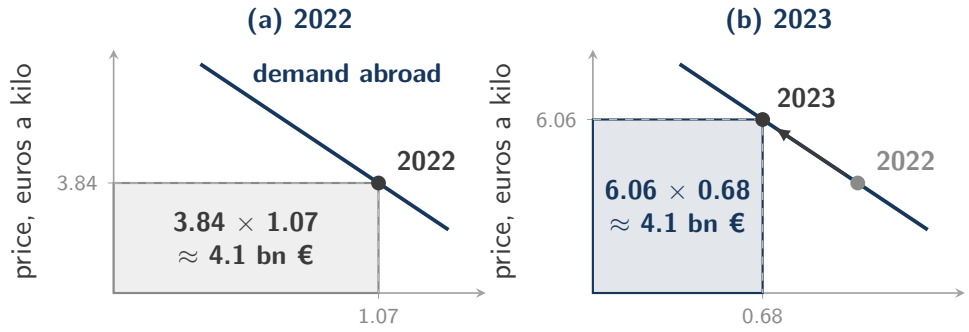
- Project: each group's question is settled today
- Not settled? See me after class
- Problem set 1 (Blocks 1–2): out today
- Due Thursday 22 October



**Slide 18** Today: your topic is fixed; Problem set 1 is out

The two practical points of the day. Each project group's question is settled today: the one agreed at the idea meeting, or by e-mail since. A group whose question is still open sees me after class, two minutes. Problem set 1 covers Blocks 1 and 2, everything up to today, and is out today; it is due on Thursday 22 October, in class. The QR code opens the course page.

## Olive oil abroad: a third less, the same money — elasticity about 1




Quantity −44 %, price +45 % (midpoint): elasticity 0.98.
Unit elastic: the price rise and the quantity fall cancel.

Spain's olive-oil exports 2022 and 2023: MAPA, Análisis del comercio exterior, aceite de oliva 2023 (customs data, AEAT; 2023 provisional). Average price = value ÷ tonnes. The curve is a sketch.



**Slide 19** Olive oil abroad: a third less, the same money: elasticity about 1

The hook's number. Value over tonnes gives the average price abroad: 3.84 euros a kilo in 2022, 6.06 in 2023. The move between the two points is up the demand curve, because what changed was the harvest, the sellers' side, exactly Class 6's olive-oil picture. By the midpoint method the quantity fell 44 per cent and the price rose 45 per cent: elasticity 0.98, as near to 1 as real data get.

Unit elastic means the price rise and the quantity fall cancel, so the two boxes, one per panel, have the same area: $3.84 \times 1.07 \approx 4.1$ billion euros in 2022, $6.06 \times 0.68 \approx 4.1$ billion in 2023 (the customs values: 4.10 and 4.15), a wide low box and a narrow tall one: a third less oil and the same money are one fact, not a paradox. Two cautions: the curve is a sketch through two points and assumes demand abroad stayed put; and the same money for the sector does not mean every mill was as well off.

## In pairs: raise the price, or cut it?

### More money, or less?

1. Rush-hour metro, elasticity 0.2: a dearer fare
   **up: few leave**
2. Cinema on a Tuesday afternoon, 2.5: a cheaper ticket
   **up: many come**
3. Bar Nou alone among twelve bars, 4.0: a dearer cup
   **down: the next cup is next door**
4. The neighbourhood's cups: 2.00 → 2.50, 160 → 120 a day
   **revenue 320 → 300: down**



## Slide 20 In pairs: raise the price, or cut it?

Exercise 2 is the slide-15 rule in use. Rush-hour metro, elasticity 0.2, inelastic: a dearer fare loses few journeys, so revenue rises; that is why fares rise and nobody stops commuting. A Tuesday-afternoon cinema, 2.5, elastic: a cheaper ticket fills seats that were empty, so revenue rises. Bar Nou alone among twelve bars, 4.0: the market is one bar's cup, narrow, with eleven substitutes in sight, so a dearer cup empties it and revenue falls; if all twelve raised together the market would be coffee on this street, less elastic.

The neighbourhood's line from 2.00 to 2.50: 320 to 300, down, because that stretch is above the middle of the line, elasticity 1.29. The trap is "a higher price always brings more money": true only below the middle.

## Income elasticity: more income, more of most goods, less of some

$$\text{income elasticity of demand} = \frac{\%\text{ change in quantity demanded}}{\%\text{ change in income}}$$

- **Normal good**: positive — a dinner out, a flat white
- **Inferior good**: negative — instant noodles, the cheapest own brand
- Among normal goods: luxuries above 1, necessities below 1

Class 5's 'income' shifter, now with a sign and a size.



## Slide 21 Income elasticity: more income, more of most goods, less of some

The same ratio with income in the denominator; this time the sign is kept, because the sign is the classification. Positive: income up, quantity up, a normal good, most things. Negative: income up, quantity down, an inferior good, the things bought because money is short, instant noodles, the cheapest own brand. Among normal goods the size sorts them: below 1 a necessity, like food as a whole, whose share of income falls as income rises; above 1 a luxury, like dinners out or concert tickets.

Class 5's income shifter moved the demand curve right or left; this number says how far.

## Cross-price elasticity: substitutes positive, complements negative

$$\text{cross-price elasticity} = \frac{\%\text{ change in quantity of A}}{\%\text{ change in price of B}}$$

- The T-usual half price → fewer T-casual journeys: **substitutes**, positive
- Croissants dearer → fewer cafès amb llet: **complements**, negative

Zero: unrelated goods. Class 5's shift of demand, measured.



**Slide 22** Cross-price elasticity: substitutes positive, complements negative

The quantity of one good over the price of another. Substitutes: the other good gets dearer and buyers move to this one, same direction, positive; Class 5's payoff was exactly this, the T-usual cheaper and T-casual journeys down. Complements: croissants dearer, fewer cafès amb llet with them, opposite directions, negative. Unrelated goods: zero.

The sign is the definition; the size says how close the substitutes are: one bar's cup against the bar next door is a large positive number, coffee against tea a small one. The usual slip is to read "both went up" as complements: look at what went up, a price and a quantity of two different goods.

## In pairs: which elasticity, what sign, what word?

### Name it, compute it, give the word.

1. Income +10 %; instant noodles −5 %
   income, **−0.5: inferior**
2. Income +10 %; concert tickets +15 %
   income, **1.5: normal, a luxury**
3. Tea +20 %; cups of coffee +4 %
   cross-price, **+0.2: substitutes**
4. Croissants +10 %; cafès amb llet −3 %
   cross-price, **−0.3: complements**
5. A cup +10 %; cups −8 %
   **own price, 0.8: inelastic** — not a cross-price



**Slide 23** In pairs: which elasticity, what sign, what word?

Exercise 3: name the elasticity first, then the sign, then the word. Income up and instant noodles down: income elasticity, negative, 0.5 in size, inferior. Income up and concert tickets up more than proportionally: income elasticity 1.5, a normal good and a luxury. Tea dearer and more cups of coffee: a price and a quantity of two different goods, cross-price, positive, 0.2, substitutes, and weak ones. Croissants dearer and fewer cafès amb llet: cross-price, negative, 0.3, complements.

Item 5 is the trap: a cup's own price and the cups sold is not a cross-price at all but the price elasticity of demand, 0.8, inelastic, and there the sign is dropped. Dropping the sign of an income or cross-price elasticity, by contrast, throws away the answer.

## Supply elasticity: sellers respond when they can; time is the key

$$\text{price elasticity of supply} = \frac{\%\text{ change in quantity supplied}}{\%\text{ change in price}}$$

- This year's olives are already on the trees: near 0
- Spare capacity, stock in the warehouse: elastic
- Years to plant, build, hire: more elastic
- Vertical: perfectly inelastic — the seats in a hall tonight

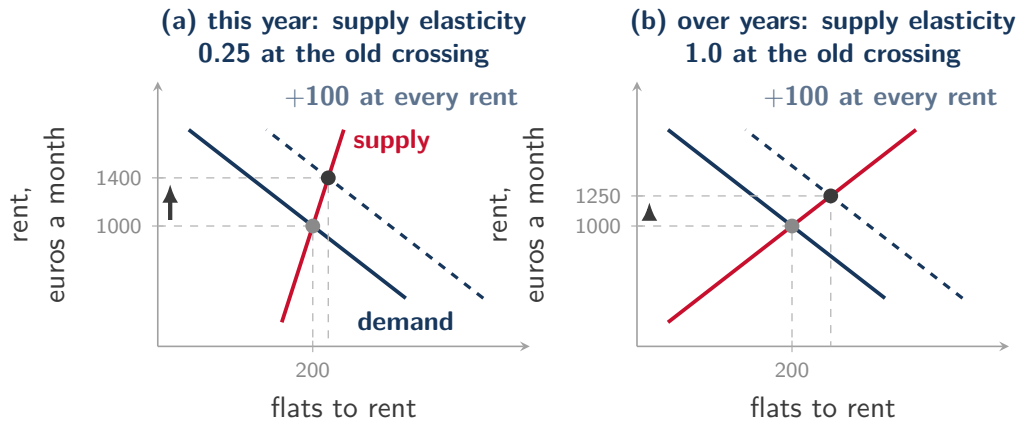


**Slide 24** Supply elasticity: sellers respond when they can; time is the key

The same ratio for the other side: how much the quantity supplied responds to the price. The reading's determinants come down to one question: can sellers change the amount, and how soon? This year's olives are already on the trees, so this year's crop cannot answer this year's price: supply near zero elasticity, which is why Class 6's olive-oil supply lines were so steep. A factory at two thirds of capacity or a full warehouse can respond at once: elastic.

Over years anyone can plant, build, hire or enter, so supply gets flatter with time. The extreme is the seats in a hall tonight: vertical, elasticity zero.

## Flats in Gràcia: more people want one; this year the rent jumps




Same shift. Steep supply: rent +400, flats +20. Flatter: rent +250, flats +50.

Own numbers; not data.

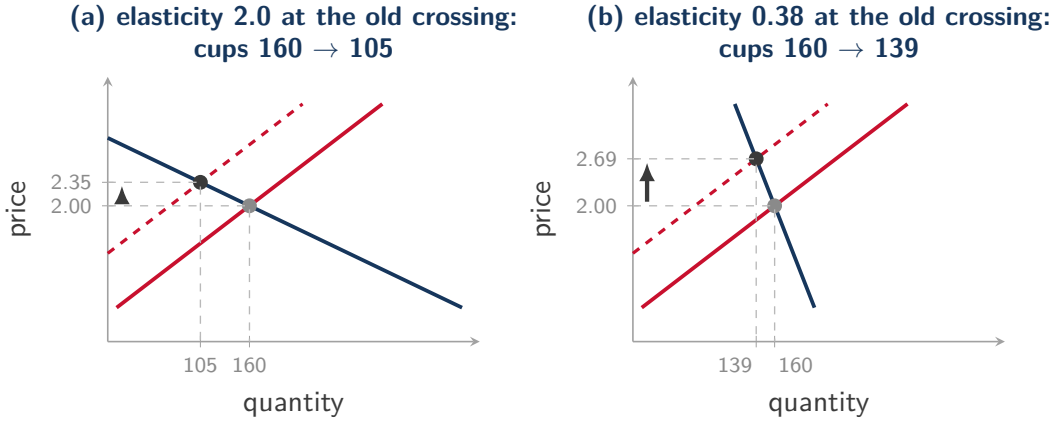


**Slide 25** Flats in Gràcia: more people want one; this year the rent jumps

Own numbers for a street in Gràcia. Demand for flats shifts right by 100 at every rent: students in September, remote workers, tourists. In (a), this year, the stock of flats is what it is and a higher rent brings only a few more out, supply elasticity 0.25 at the old crossing: the rent jumps from 1,000 to 1,400 and just 20 flats appear, so the shock goes almost entirely into the price. In (b), over years, buildings go up and owners rent out, supply elasticity 1: the same shift raises the rent to 1,250 and adds 50 flats.

The elasticity of supply decides how much of a demand shift becomes price and how much becomes quantity. When a rent jumps like this, governments cap it; what a cap does to a market with steep supply is Class 9.

## Price up 0.35 where buyers can switch, 0.69 where they cannot




Elastic buyers walk away: the quantity takes the hit. Inelastic buyers pay: the price does.



**Slide 26** Price up 0.35 where buyers can switch, 0.69 where they cannot

Slide 5 answered with numbers. The same supply shift, 90 fewer cups at every price, in both panels. In (a) demand has elasticity 2.0 at the old crossing: the price rises 0.35 and the cups fall from 160 to 105; buyers left. In (b) demand has elasticity 0.38 at the same point: the price rises 0.69, twice as far, and the cups fall only to 139; buyers stayed and paid.

Class 6's which way is the same in both panels: supply left, price up, quantity down. Today's how much is the sharing: elastic buyers put the shock into the quantity, inelastic buyers into the price. Olive oil in a Spanish kitchen is nearer (b); the bikinis would be (a).

## Two juice stalls: who should make what? (optional)

**In one hour: Red stall, 6 smoothies or 3 sandwiches; Blue stall, 2 or 2**

1. Red is faster at both. Can trade still help?
   **yes: their opportunity costs differ**
2. Who makes the sandwiches?
   **Blue: 1 smoothie each, against Red's 2**
3. A sandwich price both accept?
   **between 1 and 2 smoothies**
4. Blue's hour becomes 2 smoothies or 1 sandwich. Now?
   **no gain: now the same cost, 2 smoothies**



**Slide 27** Two juice stalls: who should make what? (optional)

Optional practice on comparative advantage, with two juice stalls, Red and Blue. Red is faster at both, but what decides who makes what is the opportunity cost: a sandwich costs Red 2 smoothies and Blue only 1, so Blue makes the sandwiches and Red the smoothies, and both gain at any price strictly between 1 and 2 smoothies a sandwich. If Blue's hour became 2 smoothies or 1 sandwich, a sandwich would cost both of them 2 smoothies: with the same opportunity costs there is nothing to gain from specialising.

## What moves on the frontier? (optional)

**A region grows almonds and makes bicycles**

1. A better almond variety doubles the harvest
   **only the almond end moves out**
2. Laid-off bicycle workers get their jobs back
   **from inside toward the frontier; the frontier stays**
3. 5,000 workers move in, able to do either job
   **the whole frontier shifts out**



### Slide 28 What moves on the frontier? (optional)

Optional practice on the production possibilities frontier. A better almond variety only helps the land used for almonds, so only the almond end of the frontier moves out. Workers getting their jobs back do not change what the region could make with everything at work: it was inside its frontier and moves toward it, and the frontier stays. New workers able to do either job move both ends out: the whole frontier shifts.

**Elasticity: the % change in quantity over the % change in price.**

**Supply shifts: demand inelastic, the price jumps; demand elastic, the quantity falls.**

Revenue follows the price when demand is inelastic, runs against it when elastic.
Sellers: steep this year, flatter with time.



### Slide 29 In one line

The class in four lines: elasticity is the percentage change in quantity over the percentage change in price; when supply shifts, inelastic demand takes the shift mostly in the price and elastic demand mostly in the quantity, which is slide 26's lesson; revenue follows the price when demand is inelastic and runs against it when elastic; supply is steep this year and flatter with time.

## Before Monday 19 October

- Block 3 begins: surplus, efficiency, taxes and subsidies
- Read Mankiw & Taylor, Ch. 6, pp. 141–158 — consumer and producer surplus, market efficiency; Ch. 7, pp. 163–171 — taxes and subsidies
- Problem set 1 (Blocks 1–2): due Thursday 22 October
- Practise in Sextant: the units the course page lists



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**Slide 30** Before Monday 19 October

Monday 19 October opens Block 3: what a market is worth to buyers and sellers, surplus and efficiency, and what taxes and subsidies do to it. Reading: Mankiw & Taylor chapter 6, pages 141 to 158, on consumer and producer surplus and market efficiency, and chapter 7, pages 163 to 171, on taxes and subsidies. Problem set 1, on Blocks 1 and 2, is due on Thursday 22 October, in class. Practise in Sextant with the units the course page lists.

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